

ROLL NO:250701573

EXPERIMENT NO: 3 HANDLING MISSING AND APPROPRIATE DATA IN A DATASET

PROGRAM: import numpy as np

import pandas as pd

df = pd.read_csv("Student_Dataset.csv")

print("ORIGINAL DATASET")

print(df)

print("\nDUPLICATE ROWS")

print(df.duplicated())

df.drop_duplicates(inplace=True)

index = np.array(list(range(len(df))))

df.set_index(index, inplace=True)

print("\nDATASET AFTER REMOVING DUPLICATES")

print(df)

df.loc[(df["Mark"] < 0) | (df["Mark"] > 100), "Mark"] = np.nan

df.loc[(df["Attendance"] < 0) | (df["Attendance"] > 100), "Attendance"] = np.nan

df.loc[df["Salary"] < 0, "Salary"] = np.nan

print("\nDATASET AFTER REPLACING INVALID VALUES")

```

print(df)

print("\nUNIQUE DEPARTMENTS")
print(df["Department"].unique())

print("\nUNIQUE PREFERENCES")
print(df["Preference"].unique())

df["Preference"] = df["Preference"].replace("coding", "Coding")
df["Preference"] = df["Preference"].replace("music", "Music")

print("\nDATASET AFTER CORRECTING VALUES")
print(df)

df["Salary"] = df["Salary"].fillna(round(df["Salary"].mean()))
df["Mark"] = df["Mark"].fillna(round(df["Mark"].median()))
df["Attendance"] = df["Attendance"].fillna(round(df["Attendance"].median()))
df["Preference"] = df["Preference"].fillna(df["Preference"].mode()[0])

print("\nFINAL CLEANED DATASET")
print(df)

```

OUTPUT:

ORIGINAL DATASET

	StudentID	Age	Department	Mark	Attendance	Salary	Preference
0	1	19	CSE	85	90	30000	Coding
1	2	20	IT	78	85	32000	Sports
2	3	19	CSE	105	88	30000	coding
3	4	21	ECE	-1	75	35000	Music
4	5	20	IT	65	-5	33000	Coding

5	6	19	CSE	92	95	-1000	Sports
6	7	21	ECE	75	80	36000	music
7	7	21	ECE	75	80	36000	music
8	8	20	IT	88	92	34000	Coding
9	9	19	CSE	80	89	31000	Sports

DUPLICATE ROWS

```

0    False
1    False
2    False
3    False
4    False
5    False
6    False
7     True
8    False
9    False
dtype: bool

```

DATASET AFTER REMOVING DUPLICATES

	StudentID	Age	Department	Mark	Attendance	Salary	Preference
0	1	19	CSE	85	90	30000	Coding
1	2	20	IT	78	85	32000	Sports
2	3	19	CSE	105	88	30000	coding
3	4	21	ECE	-1	75	35000	Music
4	5	20	IT	65	-5	33000	Coding
5	6	19	CSE	92	95	-1000	Sports
6	7	21	ECE	75	80	36000	music
7	8	20	IT	88	92	34000	Coding
8	9	19	CSE	80	89	31000	Sports

DATASET AFTER REPLACING INVALID VALUES

	StudentID	Age	Department	Mark	Attendance	Salary	Preference
0	1	19	CSE	85.0	90.0	30000.0	Coding
1	2	20	IT	78.0	85.0	32000.0	Sports
2	3	19	CSE	NaN	88.0	30000.0	coding
3	4	21	ECE	NaN	75.0	35000.0	Music
4	5	20	IT	65.0	NaN	33000.0	Coding
5	6	19	CSE	92.0	95.0	NaN	Sports
6	7	21	ECE	75.0	80.0	36000.0	music
7	8	20	IT	88.0	92.0	34000.0	Coding
8	9	19	CSE	80.0	89.0	31000.0	Sports

UNIQUE DEPARTMENTS

```

<StringArray>
['CSE', 'IT', 'ECE']
Length: 3, dtype: str

```

UNIQUE PREFERENCES

```
<StringArray>
['Coding', 'Sports', 'coding', 'Music', 'music']
Length: 5, dtype: str
```

DATASET AFTER CORRECTING VALUES

	StudentID	Age	Department	Mark	Attendance	Salary	Preference
0	1	19	CSE	85.0	90.0	30000.0	Coding
1	2	20	IT	78.0	85.0	32000.0	Sports
2	3	19	CSE	NaN	88.0	30000.0	Coding
3	4	21	ECE	NaN	75.0	35000.0	Music
4	5	20	IT	65.0	NaN	33000.0	Coding
5	6	19	CSE	92.0	95.0	NaN	Sports
6	7	21	ECE	75.0	80.0	36000.0	Music
7	8	20	IT	88.0	92.0	34000.0	Coding
8	9	19	CSE	80.0	89.0	31000.0	Sports

FINAL CLEANED DATASET

	StudentID	Age	Department	Mark	Attendance	Salary	Preference
0	1	19	CSE	85.0	90.0	30000.0	Coding
1	2	20	IT	78.0	85.0	32000.0	Sports
2	3	19	CSE	80.0	88.0	30000.0	Coding
3	4	21	ECE	80.0	75.0	35000.0	Music
4	5	20	IT	65.0	88.0	33000.0	Coding
5	6	19	CSE	92.0	95.0	32625.0	Sports
6	7	21	ECE	75.0	80.0	36000.0	Music
7	8	20	IT	88.0	92.0	34000.0	Coding
8	9	19	CSE	80.0	89.0	31000.0	Sports